Software Engineering

Tahir Farooq
FAST-NU

WHY SOFTWARE FAILURES?

WHAT ARE CAUSES?

Deficiencies in software quality often results

- In costly emergency fixes
- Damage to a brand’s reputation, software company defaulters–large repayments

Software defects are extremely costly in term of

- Money
- Loss of life
- Reputation

Software Failures

Formula written on paper was improperly transcribed

NASA: Mariner Failure

- A bug in the flight software for the Mariner 1
- The rocket to **divert from its intended path**
- Mission control destroys the rocket over the Atlantic Ocean
- The investigation discovers that a formula written on paper was improperly transcribed

Software Failures

<u>**Rocket Launch Errors**</u>

- In 1996, a European Ariane5 rocket was launched rocket to veer off its path a mere 37 seconds after launch
- **As it started disintegrating, it self-destructed**

- The problem was the result of code reuse from the launch system's predecessor
- More than $370 million were lost due to this error

Software Failures

Communication b/w software and hardware error

Flight Crashes

In 1994 in Scotland, a Chinook helicopter crashed

- ❑ killed all 29 passengers

- ❑ The crash was due to a systems error

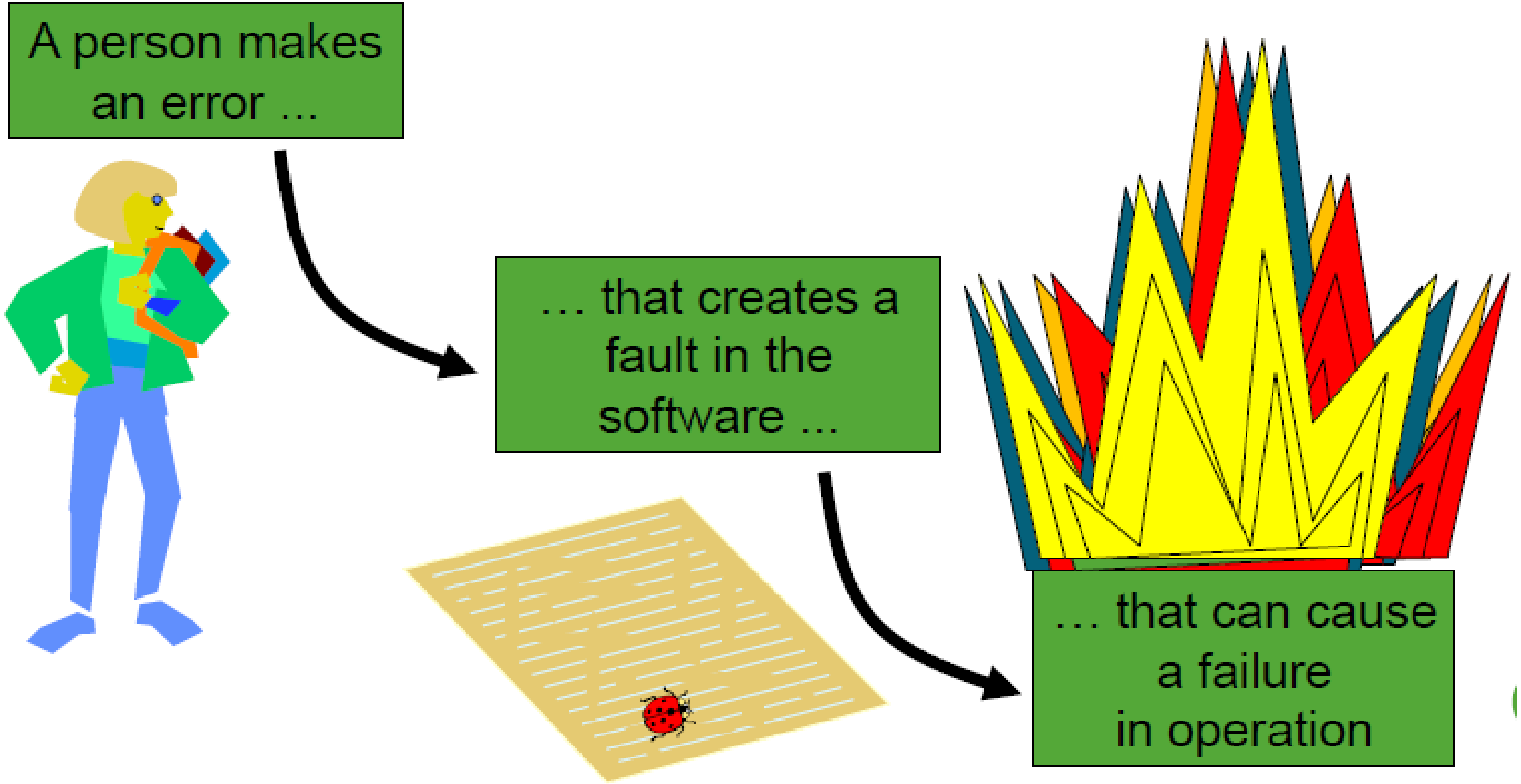
Software Failures

<u>Korean Airliner Crash</u>

- ❑ KAL 801 got accident
- ❑ Killed 225 out of 254 aboard

- ❑ A software design problem was discovered in barometric altimetry in Ground Proximity Warning System (GPWS)

Error – Fault – Failure

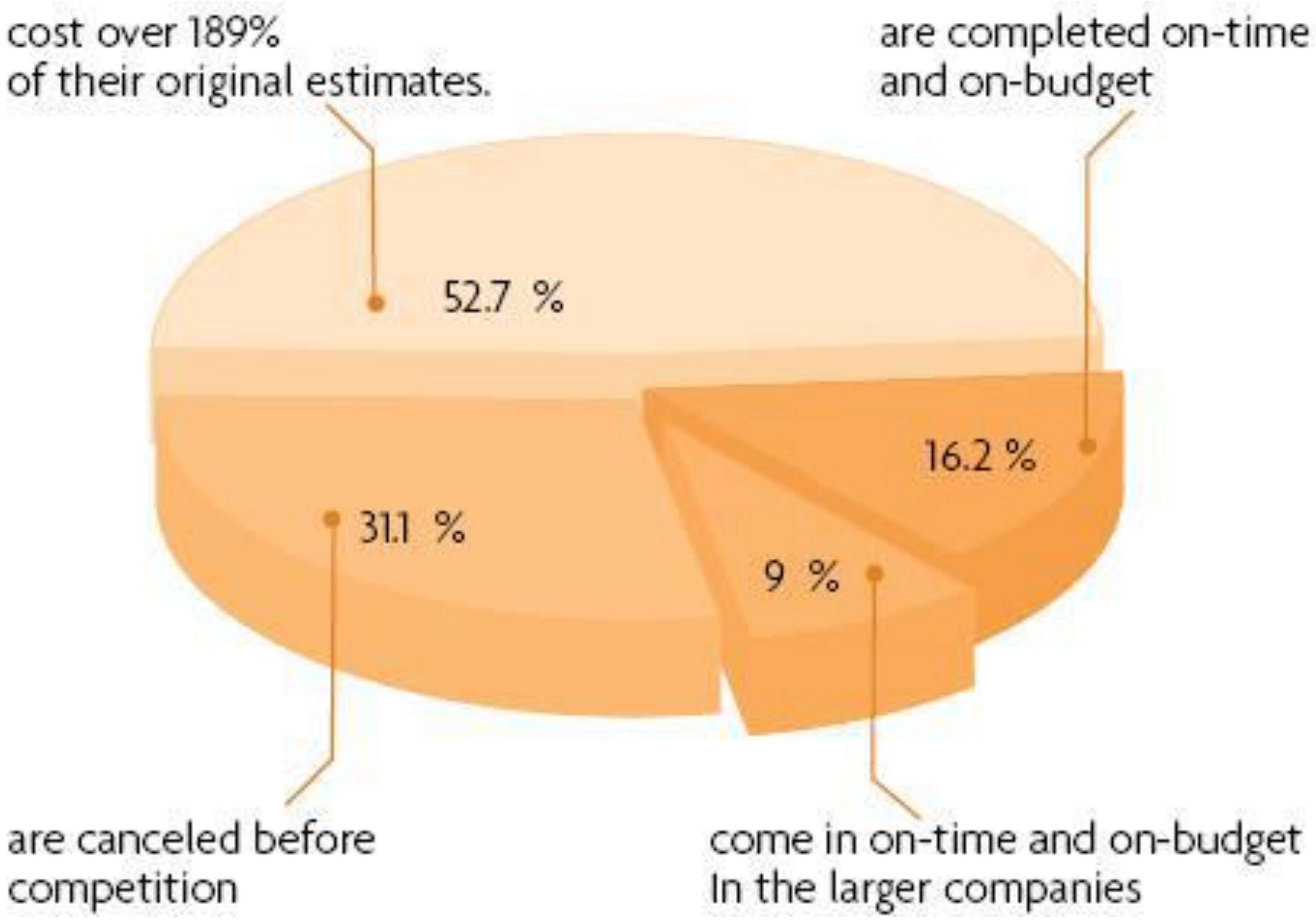


Some Famous Software Errors

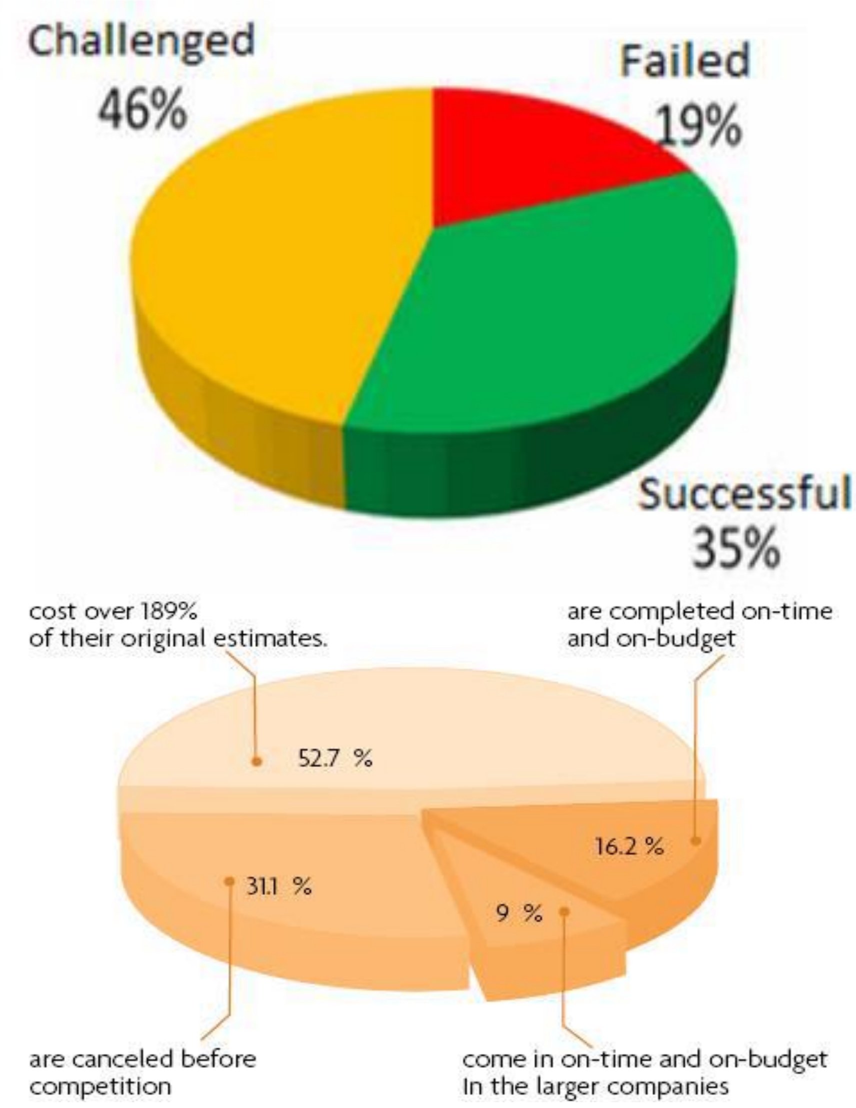
Announcement

- Therac-25
- Patriot Missile System
- NASA's Mars Polar Lander
- ESA's Ariane5 Launch System
- 2003 Blackout

Standish Group Chaos Report



Standish Group Chaos Report (2007)



- ❑ 35%of all software projects are now completed successful (1995: 16.2 %)
- ❑ 19% of projects fail (1995: 31.1%)
- ❑ 46%where described as challenged, meaning they had cost or time overruns or didn't fully meet the user's needs (1995: 52.7%)

Importance of SQA

- Due to the **widespread acceptance**, and use of software systems software bugs are proving to be costly, and sometimes fatal.

- It is **estimated** that **buggy** or **flawed** software cost businesses $175 billion worldwide in 2001.

- Bugs have affected banking systems, stock exchanges, medical institutions, educational institutions and even the Social Security Administration

Importance of SQA

Most bugs, **encountered** during **software development**,

can be avoided,

by adopting a sound software **development process**,

and

having strict **software quality control** using Software Quality Assurance

Software Quality Assurance

VS

Software Testing

SQA vs. Software Testing

- Software Testing involves **operating** a system, or an application, under **controlled conditions**, and **evaluating the results.**

- Software testing is normally carried out under controlled conditions

- The **aim of testing** is to try to **break the software**, and find the **bugs** in it.
- Testing is oriented towards "**detection** "of bugs in the software

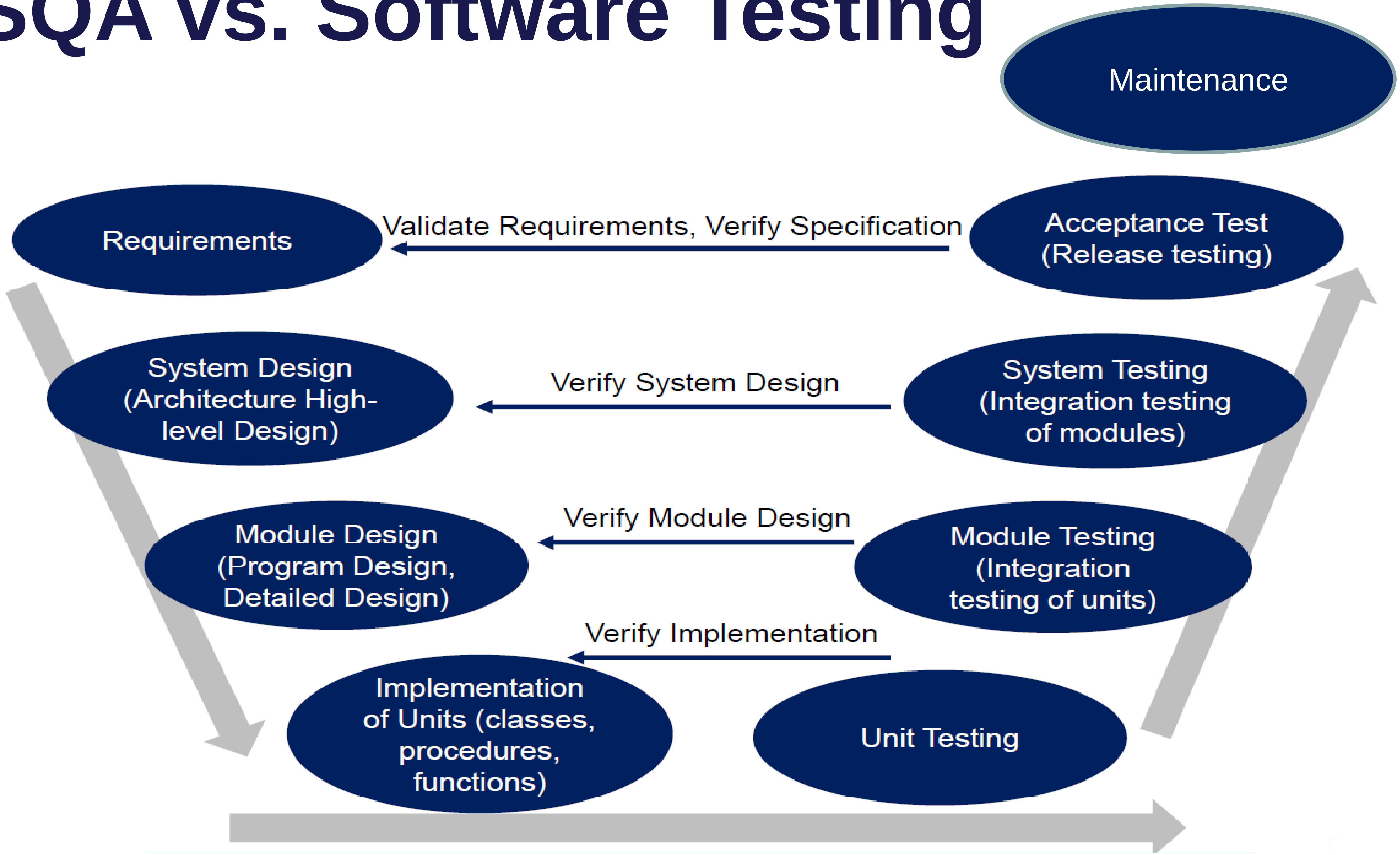
SQA vs. Software Testing

- SQA is aimed at **Avoiding Bugs.**

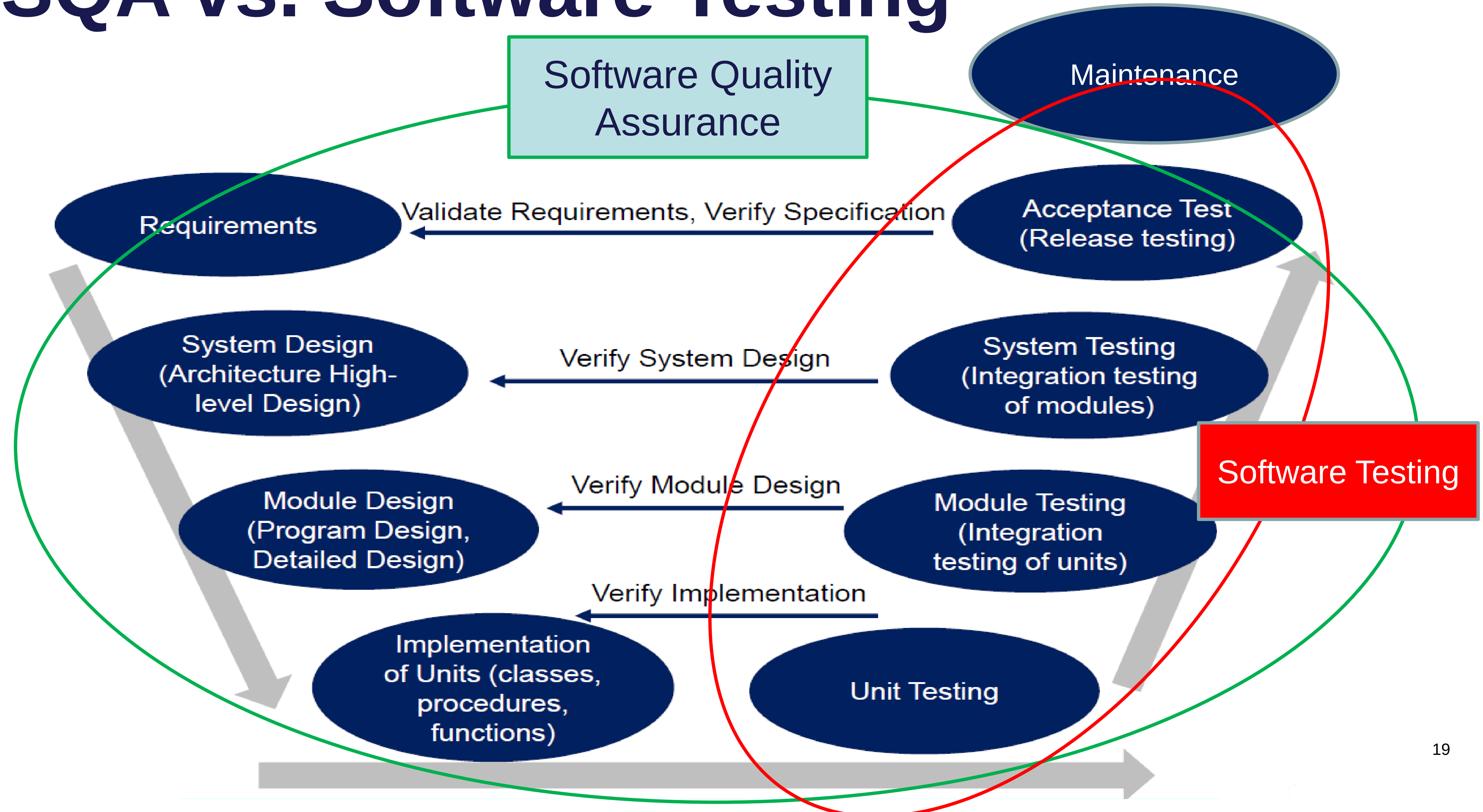
- Software Quality Assurance is oriented towards "**prevention** "of bugs in the software, by following a software development methodology.

- SQA is more concerned with developing a quality process for software development, which will prevent the generation of bugs, and will result in the production of quality software

SQA vs. Software Testing



SQA vs. Software Testing



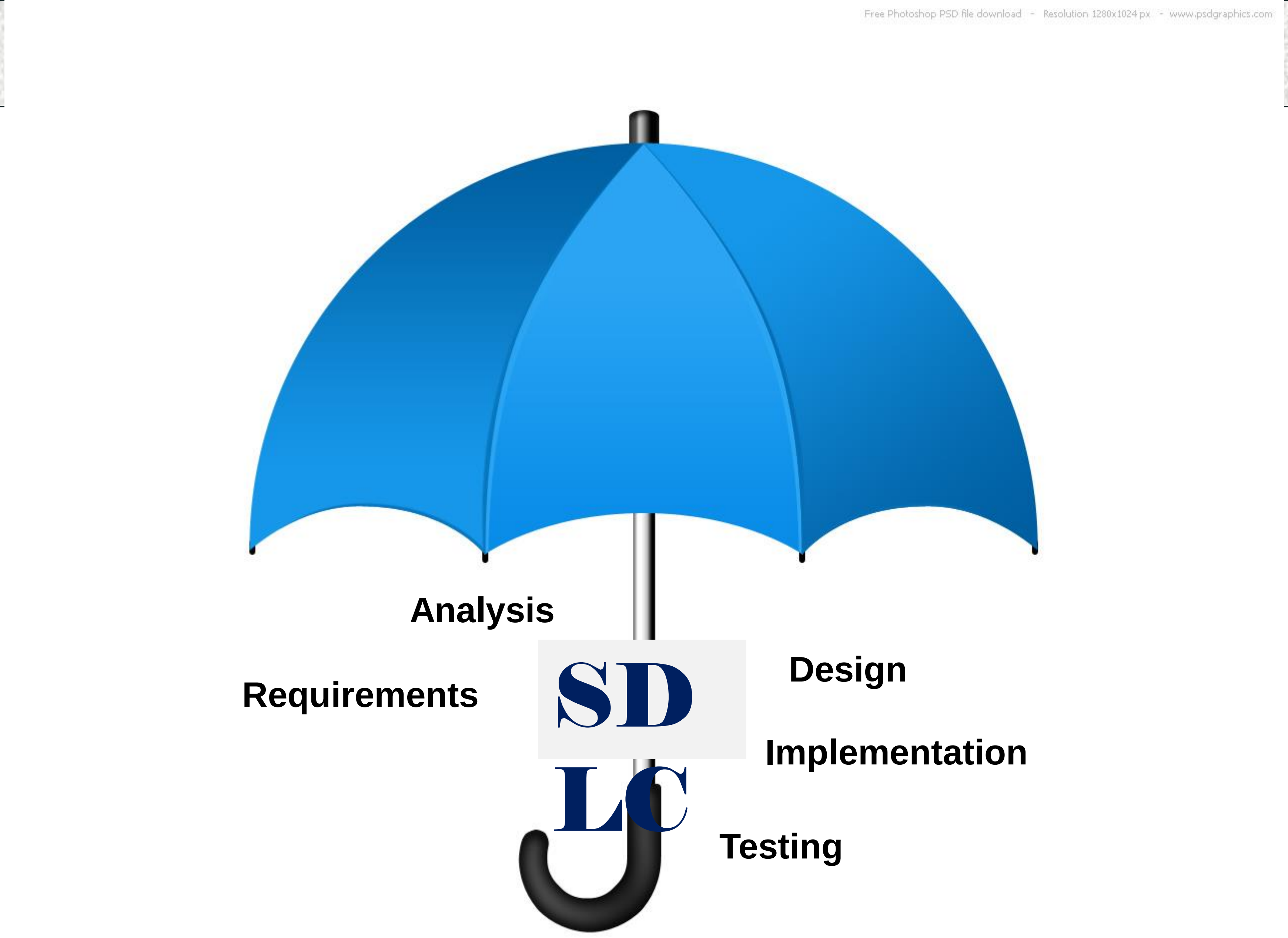
What is Quality

- In general, people's quality expectations for software system they use and rely upon are two-fold:

1. The software systems must do what they are supposed to do. In other words, they must **do the right things**.

2. They must perform these specific tasks correctly or satisfactorily. In other words, they must **do the things right.**

Now you can define Software Quality Assurance.


Free Photoshop PSD file download - Resolution 1280x1024 px - www.psdgraphics.com
Analysis
Requirements
SD
LC
Design
Implementation
Testing

What is SQA?

- Software Quality Assurance is an **umbrella activity** that is applied throughout the software process...

What is quality?

- *Quality* refers to any measurable characteristics such as **correctness**, **maintainability**, **portability**, **testability**, **usability**, **reliability**, **efficiency**, **integrity**, **reusability** and **interoperability**.

Quality terminologies

- ***Quality of Design*** refers to the characteristics that designer's specify for an item.

- ***Quality of Conformance*** is the degree to which the design specifications are followed during manufacturing.

- ***Quality Control*** is the series of inspections, reviews and tests used throughout the development cycle to ensure that each work product meets the requirements placed upon it.

- ***Quality policy*** refers to the basic aims and objectives of an organization regarding quality as stipulated by the management.

Quality terminologies

Quality assurance consists of the auditing and reporting functions of management.

Cost of Quality includes all costs incurred in the pursuit of quality or in performing quality related activities such as appraisal costs, failure costs and external failure costs.

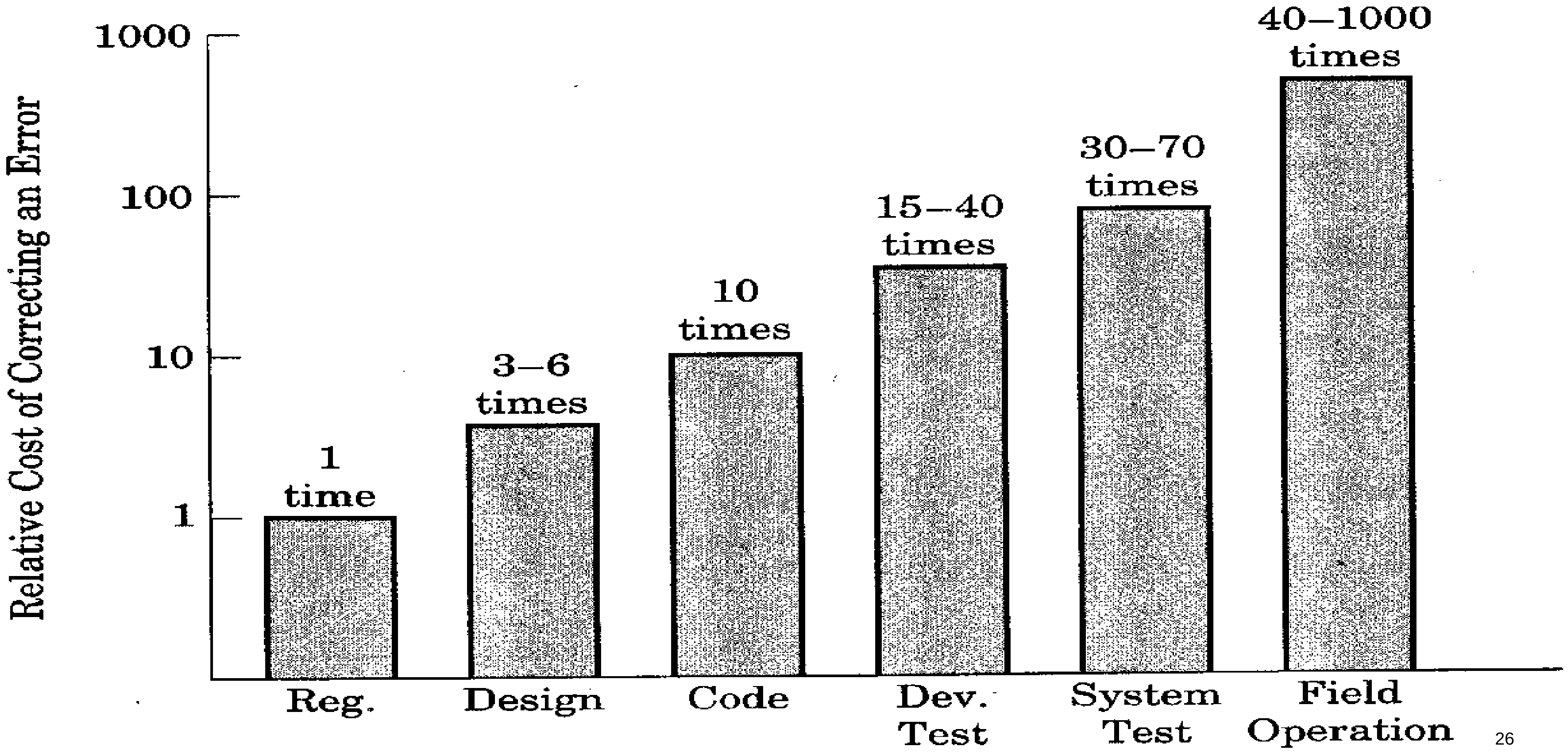
Quality planning is the process of assessing the requirements of the procedure and of the product and the context in which these must be observed.

Quality testing is assessment of the extent to which a test object meets given requirements

Quality assurance plan is the central aid for planning and checking the quality assurance.

Quality assurance system is the organizational structure, responsibilities, procedures, processes and resources for implementing quality management.

Relative cost of correcting an error?



Elements of S/W Quality Assurance

- *Standards (ISO 9001, CMMI)*
- *Reviews and audits*
- *Testing*
- *Error/defect collection and analysis*
- *Change management*
- *Education*
- *Vendor management*
- *Security management*
- *Safety*
- *Risk management*

SQA tasks

- *Prepares an SQA plan for a project*
- *Participates in the development of the project's software process description*
- *Reviews software engineering activities to verify compliance with the defined software process*
- *Audits designated software work products to verify compliance with those defined as part of the software process*
- *Ensures that deviations in software work and work products are documented and handled according to a documented procedure*
- *Records any noncompliance and reports to senior management*